

# SUDIPTA MONDAL

Maryland, USA

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in [LinkedIn](#) [GitHub](#) [Google Scholar](#) [Portfolio Webpage](#)

## Summary

Ph.D. student in Computer Science at the University of Maryland, Baltimore County, specializing in HCI and NLP. Passionate about advancing AI through interdisciplinary research, with the goal of driving impactful innovation in intelligent systems and pursuing a career in academia or industry research to mentor and inspire the next generation of scholars.

## Academic Preparation

### University of Maryland, Baltimore County

Fall 2025 – Present

*Doctor of Philosophy (Ph.D.), Computer Science*

**Research Focus : HCI, NLP, AI in Education**

MD, USA

### University of Maryland, Baltimore County

Fall 2025 – Spring 2027

*Concurrent - M.S. in Computer Science*

**CGPA: 3.925/4.00**

MD, USA

### Brac University

2016 – 2021

*Bachelor of Science in Computer Science and Engineering*

**CGPA: 3.49/4.00** (last 60 credits: 3.61)

Bangladesh

## Experience

### Graduate Teaching Assistant

Aug 2025 – Present

*Department of Computer Science and Electrical Engineering, UMBC*

**Course:** CMSC331 - Principles of Programming Languages

**Responsibilities:** Grading assignments, homework, and tests; holding office hours for student consultation; proctoring exams

## Research Interest

**Human-Centered AI** | Human Computer Interaction, Natural Language Processing, AI in Education

## Notable Publications

- Anan, R., Apon, T. S., Hossain, Z. T., Modhu, E. A., **Mondal, S.**, & Alam, M. G. R. (2023, March). *Interpretable Bangla Sarcasm Detection Using BERT and Explainable AI*. In *2023 IEEE 13th Annual Computing and Communication Workshop and Conference (CCWC)* (pp. 1272–1278). IEEE. DOI: [10.1109/CCWC57344.2023.10099331](https://doi.org/10.1109/CCWC57344.2023.10099331)
- Dohan, D. M., Prottush, N., **Mondal, S.**, Choya, S. B. Z., & Alam, M. G. R. (2025, April). *Customer Personality Analysis Using Machine Learning with Explainable AI*. In *2025 13th International Symposium on Digital Forensics and Security (ISDFS)* (pp. 1–6). IEEE. DOI: [10.1109/ISDFS65363.2025.11012007](https://doi.org/10.1109/ISDFS65363.2025.11012007)
- Islam, A., Rahman, S. A. S., Bhowmick, P., Rafi, I. A., **Mondal, S.**, & Alam, G. R. (2025, January). *A Smart Tutor Avatar for Mimicking Characters of Bangla Sign Language*. In *2025 4th International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST)* (pp. 183–188). IEEE. DOI: [10.1109/ICREST63960.2025.10914355](https://doi.org/10.1109/ICREST63960.2025.10914355)
- Islam, A., Rafi, I. A., **Mondal, S.**, Rahman, S. A. S., & Alam, G. R. (2025, December). *Which Matters More: Model or Language? An Empirical Study in English-Bangla Mental Health Classification*. In *2025 5th International Conference on Robotics, Automation, and Artificial Intelligence (RAAI)* (pp. 114–118). IEEE. DOI: [10.1109/RAAI67517.2025.11423350](https://doi.org/10.1109/RAAI67517.2025.11423350)
- Islam, A., Rafi, I. A., Rahman, S. A. S., **Mondal, S.**, & Alam, M. G. R. (2025, December). *Rice Leaf Disease Classification: A Comparative Evaluation of CNNs and Vision Transformers*. In *2025 28th International Conference on Computer and Information Technology (ICCIT)* (pp. 5537–5542). IEEE. DOI: [10.1109/ICCIT68739.2025.11490559](https://doi.org/10.1109/ICCIT68739.2025.11490559)

## Ongoing works

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- Studying career decision-making challenges among undergraduate computer science students through a mixed-methods approach combining a validated assessment with reflective prompts.
- Analyzing student academic advising queries to uncover patterns and information gaps, toward designing an AI-assisted tool for faculty advisors.

## Notable Course Projects

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### Classify Interpretable Twitter Hate Speech using LIME | *CSE711: Symbolic Machine Learning-I*

- [\[Project Link\]](#): Conducted a comparative performance analysis of multiple machine learning models and incorporated explainability techniques using LIME for better interpretability.

### Deep Learning Based Online Sexism Detection | *CSE712: Symbolic Machine Learning-II*

- [\[Project Link\]](#): Developed a neural network using Word2Vec embeddings to detect and categorize sexist language, achieving F1 scores of 0.77 for detection and 0.43 for categorization.

### Social Media Rumor Detection using Machine Learning | *CSE718: Petri Net Theory and Modeling of Systems*

- [\[Project Link\]](#): Built a machine learning system to detect social media rumors by applying two feature extraction methods for data vectorization, training multiple classification models, and identifying SVM as the top-performing model.

## Technical Skills

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**Language:** Python, Java

**Developer Tools:** PyCharm, Jupyter Notebook, Google Colab, VS Code

**Operating Systems:** Linux, Windows

**Machine Learning Frameworks:** Tensorflow, Scikit-learn, Keras, PyTorch

**Research Methods:** Qualitative Coding of User Study Data, Mixed-Methods Analysis

**Others:** Git, SQL, Latex

## Certificates & Awards

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### VC's List and Dean's List, Brac University

- Earned a position on the **VC's List** for the Spring 2020 semester with a 4.00 GPA
- Earned a place on the **Dean's List** for two semesters with a 3.80 and 3.87 GPA

## Recommendation

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**Sanorita Dey, PhD**

*Assistant Professor*

Dept. of CSEE, University of Maryland, Baltimore County

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